4. Which one of the following is an example of polygenic inheritance?
   a) Flower colour in *Mirabilis Jalapa*
   b) Production of male honey bee
   c) Pod shape in garden pea
   d) Skin Colour in humans

5. In Mendel's experiments with garden pea, round seed shape (RR) was dominant over wrinkled seeds (rr), yellow cotyledon (YY) was dominant over green cotyledon (yy). What are the expected phenotypes in the F_2 generation of the cross RRYY x rryy?
   a) Only round seeds with green cotyledons
   b) Only wrinkled seeds with yellow cotyledons
   c) Only wrinkled seeds with green cotyledons
   d) Round seeds with yellow cotyledons an wrinkled seeds with yellow cotyledons

6. Test cross involves
   a) Crossing between two genotypes with recessive trait
   b) Crossing between two F_1 hybrids
   c) Crossing the F_1 hybrid with a double recessive genotype
   d) Crossing between two genotypes with dominant trait

7. In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seed pant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F_1 generation?
   a) 9:1 b) 1:3
   b) 3:1 d) 50:50

8. The genotype of a plant showing the dominant phenotype can be determined by
   a) Back cross b) Test cross
   c) Dihybrid corss d) Pedigree analysis

9. Select the correct statement from the ones given below with respect to dihydrid cross
   a) Tightly linked genes on the same chromosomes show very few combinations
   b) Tightly linked genes on the same chromosomes show higher combinations
   c) Genes far apart on the same chromosomes show very few recombinations
   d) Genes loosely linked on the same chromosomes show similar recombinations as the tightly linked ones

10. Which Mendelian idea is depicted by a cross in which the F_1 generation resembles both the parents
    a) Incomplete dominance
    b) Law of dominance
    c) Inheritance of one gene
    d) Co-dominance

11. Fruit colour in squash is an example of
    a) Recessive epistatsis
    b) Dominant epistasis
    c) Complementary genes
    d) Inhibitory genes

12. In his classic experiments on Pea plants, Mendel did not use
    a) Flowering position b) Seed colour
    c) Pod length d) Seed shape

13. The epistatic effect, in which the dihybrid cross 9:3:3:1 between AaBb Aabb is modified as
    a) Dominance of one allele on another allele of both loci
    b) Interaction between two alleles of different loci
    c) Dominance of one allele to another alleles of same loci
    d) Interaction between two alleles of some loci

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